

INDIAN SHOPPING MELA

Developer Architecture Master Plan

Marketplace Platform • Seller • Catalogue • Bulk Upload • Video • Orders • Shipping • Returns • Payouts • Admin

PURPOSE: This document is the implementation baseline. Anything marked **MUST** is required for launch unless formally moved to a later phase in writing. The developer must not assume a feature is optional because it is not visible on the customer website.

Document Control

Field	Value
Document	Indian Shopping Mela Developer Architecture Master Plan
Version	V1
Platform Type	Australia-wide multi-vendor e-commerce marketplace
Primary Actors	Customer, Seller, Admin, Support, Courier/Shipping API, Payment Provider
Payout Rule	Seller payout eligible 14 days after confirmed delivery, subject to holds
Return Rule	7-day ordinary change-of-mind window from confirmed delivery; statutory consumer rights handled separately
Priority Requirement	Seller listing, BULK PRODUCT UPLOAD and product VIDEO support are launch-critical

Implementation Language

- **MUST** = launch-critical requirement.
- **SHOULD** = strongly recommended and should be designed now even if activated later.
- **MAY** = optional or future enhancement.
- No destructive business-rule change may be hard-coded where an admin-configurable setting is reasonable.

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1. Scope & Non-Negotiables

The platform must be designed as a **TRUE MULTI-VENDOR MARKETPLACE**, not a normal e-commerce store with multiple suppliers.

Requirement	Launch Priority	Developer Rule
Multi-vendor seller accounts	MUST	Independent seller identity, store, products, orders, returns and payouts.
Single product listing	MUST	Full structured listing with variants, inventory, shipping data and media.

Requirement	Launch Priority	Developer Rule
Bulk product upload	MUST	CSV/XLSX import with validation, preview, error report and retry.
Product video	MUST	Seller can add approved product videos without breaking page performance.
Multi-seller cart	MUST	One customer checkout; backend splits into seller sub-orders.
Shipping integration	MUST	API-ready label, tracking and delivery-state architecture.
Returns	MUST	Return workflow connected to payout holds.
Seller payout	MUST	Eligible 14 days after confirmed delivery unless on hold.
Audit trail	MUST	Important status and financial changes logged with actor + timestamp.
Admin configuration	MUST	Commission, payout delay, return window, listing rules and seller SLAs configurable.

Architecture Guardrails

- Never calculate seller payout from the customer-facing order total alone. Use an immutable transaction/ledger breakdown.
- Never let a seller see another seller's customer/order/commercial information.
- Never let stock become negative because two customers purchased the same last unit concurrently.
- Never rely on a manual tracking number when the integrated shipping provider supplies one.
- Never delete an order, payout, refund or transaction record. Use status changes and audit history.
- Never allow a bulk upload to publish malformed products silently. Import must validate first.
- Never store video/media in a way that makes product pages slow or unsafe.

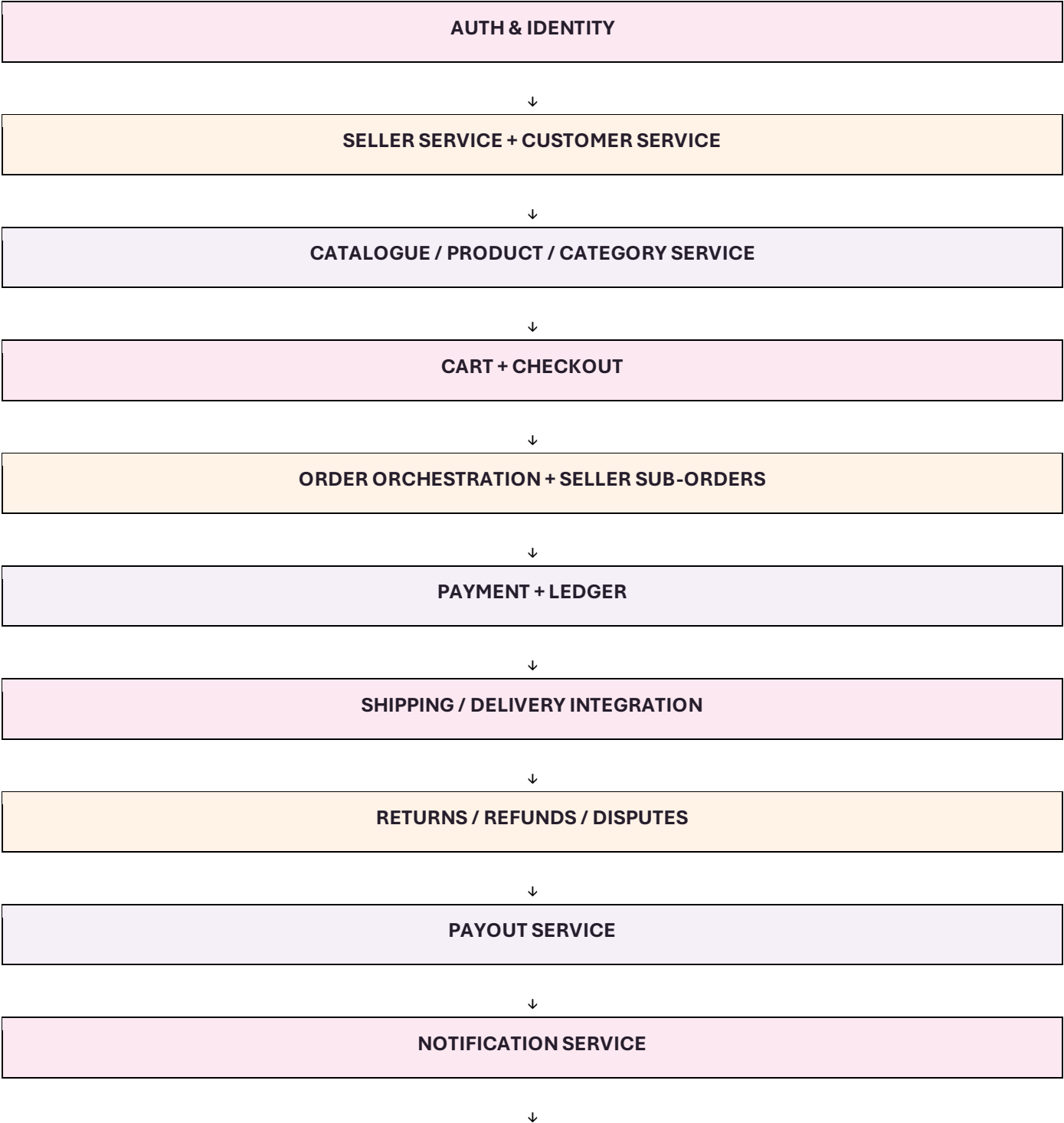
2. Roles & Permissions

Role	Core Access	Restrictions
Guest Customer	Browse, search, view product, cart	Cannot access account/order details until authenticated or verified guest flow.
Registered Customer	Checkout, orders, returns, reviews, addresses, wishlist	Only own orders/data.
Seller Owner	Store, products, inventory, orders, shipping, returns, payouts, team	Only own seller account.
Seller Staff	Permission-based seller functions	No bank/payout access unless granted.
ISM Support	Orders, tickets, returns, customer/seller assistance	No sensitive payout settings unless specifically authorised.
ISM Catalogue Admin	Categories, product moderation, attributes, bulk import review	No bank changes by default.
ISM Finance Admin	Payouts, refunds, financial reconciliation	Restricted and audited.

Role	Core Access	Restrictions
ISM Super Admin	All platform settings	Strong MFA and complete audit logging.

Seller team permissions SHOULD support: Products, Inventory, Orders, Shipping, Returns, Promotions, Reports, Store Settings and Finance as separate toggles.

3. Core System Architecture



SEARCH / INDEXING



ADMIN / REPORTING / AUDIT

The developer may implement this as a modular monolith or services, but the domain boundaries above must remain clear so the platform can scale and integrations can be replaced later.

4. Seller Onboarding — Detailed

1. Seller clicks Become a Seller.
2. Create account with email + mobile and verify both.
3. Select seller type: Individual/Sole Trader, Company/Business, Other permitted type.
4. Enter legal/business name, trading name, ABN, contact person and customer-facing store name.
5. Enter dispatch address and return address. Allow same-as-dispatch toggle.
6. Enter bank/payout details through the selected marketplace payment provider wherever possible.
7. Upload required verification documents if required by business rules/provider.
8. Select selling categories. Flag restricted categories for manual approval.
9. Accept Seller Agreement, commission, shipping, returns, prohibited products and payout terms.
10. Submit application.
11. System assigns onboarding status: DRAFT → SUBMITTED → UNDER_REVIEW → INFO_REQUIRED → APPROVED / REJECTED / SUSPENDED.
12. Admin reviews seller and records internal notes.
13. On approval, seller store and dashboard are activated.
14. Seller receives onboarding checklist: complete store, set shipping defaults, add first product, add payout details, test listing.

Field Group	Minimum Fields
Identity	Seller ID, legal name, trading/store name, contact name, email, mobile
Business	ABN, business type, category selection
Addresses	Dispatch, return, suburb, state, postcode, country
Payout	Provider account ID / verified payout destination; never expose full sensitive account info unnecessarily
Policy	Terms version accepted, timestamp, IP/device metadata where appropriate
Status	Onboarding status, risk flag, approval date, approving admin

5. Seller Storefront

- MUST have unique seller slug/URL and seller ID.
- Logo, banner, store name, About, dispatch location, seller rating and policies.
- Seller products with category/filter support.
- Store status: Draft, Active, Holiday Mode, Suspended, Closed.
- Holiday Mode must stop/limit new orders without deleting products.
- Seller can define handling/dispatch time defaults; product-specific value may override.
- Admin can suspend store or individual products without deleting historical order records.

6. Catalogue & Category Architecture

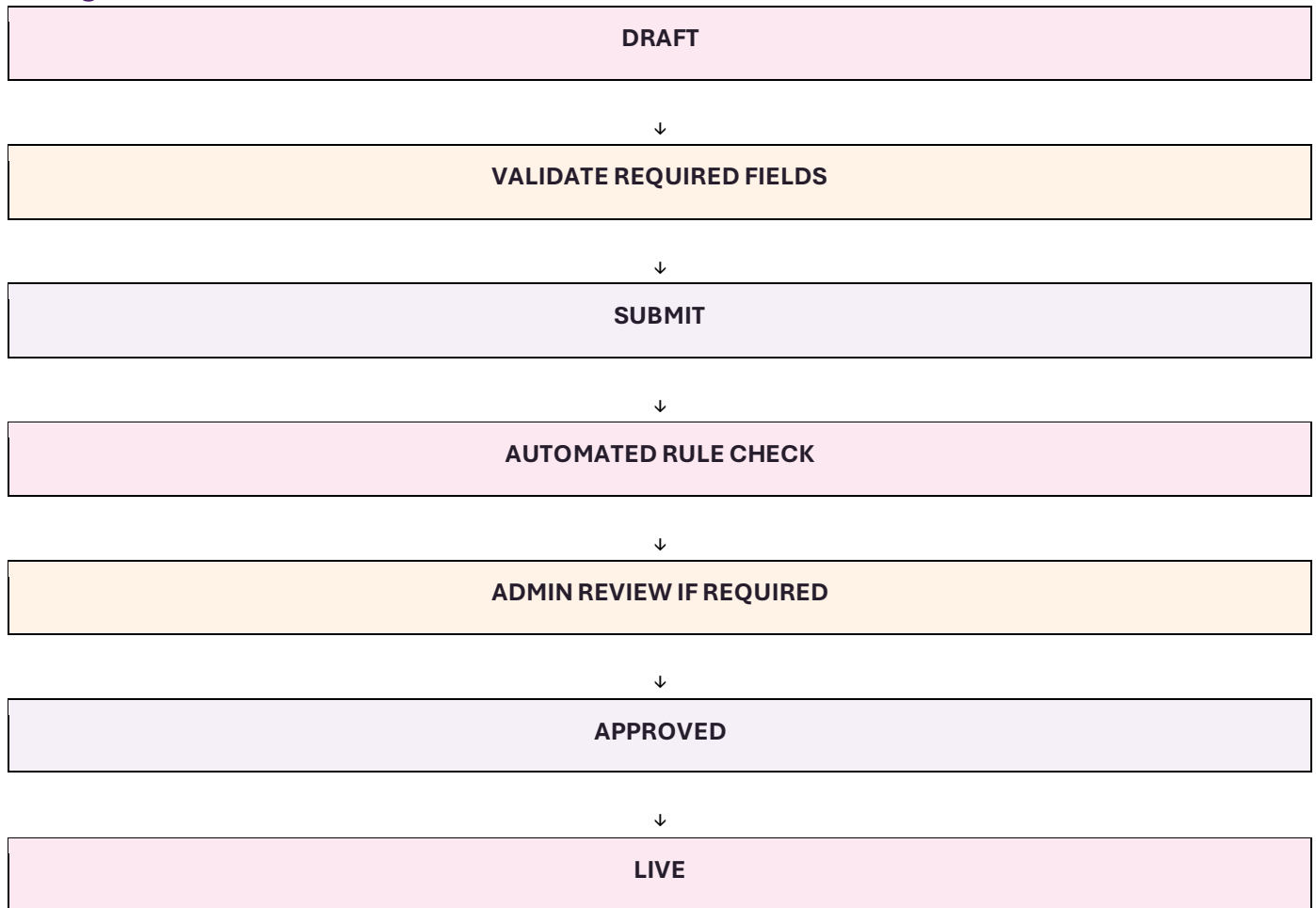
- Categories must support unlimited practical hierarchy: Department → Category → Subcategory.
- Each category can have category-specific attributes. Example: saree fabric; footwear size; bedsheet size; jewellery material.
- Attributes must be configurable by Admin, not hard-coded in product form.
- Attribute types: text, number, decimal, dropdown, multi-select, colour, size, boolean, date and dimension.
- Categories support SEO title, description, image/banner, active/inactive, sort order, featured flag and festival/occasion tags.
- Products may belong to one primary category and multiple discovery collections/tags.

Catalogue Object	Examples
Department	Women, Men, Kids, Jewellery, Home & Living, Pooja, Festivals, Gifts
Category	Sarees, Bedsheets, Footwear, Diyas
Subcategory	Kanjivaram Sarees, Jaipuri Bedsheets, Women's Juttis
Attribute	Fabric, Size, Colour, Material, Occasion
Collection	Diwali Edit, Wedding Shop, Under \$50, New Arrivals
Tag	Handmade, Ready to Ship, Australia Stock, Giftable

7. Single Product Listing — Launch-Critical

Section	Required Capability
Identity	Product title, seller SKU, auto product ID, brand/maker optional
Classification	Department, category, subcategory, occasion/festival, tags
Description	Short description, full description, key features, care instructions
Media	Primary image, gallery images, optional/allowed videos
Price	Regular price, sale price, sale dates, compare-at price if used
Inventory	Stock quantity, low-stock threshold, stock status, backorder setting if allowed
Variants	Size, colour, material/style etc. Each sellable combination can have SKU/price/stock.
Shipping	Product weight, package weight, dimensions, handling time, fragile flag
Returns	Return eligible yes/no subject to policy, category restrictions, hygiene/custom flags
Compliance	Country of origin where required, warnings/instructions where relevant
Status	Draft, Submitted, Needs Changes, Approved, Live, Paused, Out of Stock, Rejected, Archived

Listing workflow



Editing a live product must not erase order-history data. Material changes MAY trigger re-review based on configurable rules.

8. BULK PRODUCT UPLOAD — CRITICAL / MUST NOT BE OMITTED

Bulk upload is a launch requirement. Seller must be able to onboard tens, hundreds or thousands of products without manually creating every listing.

8.1 Supported Import Methods

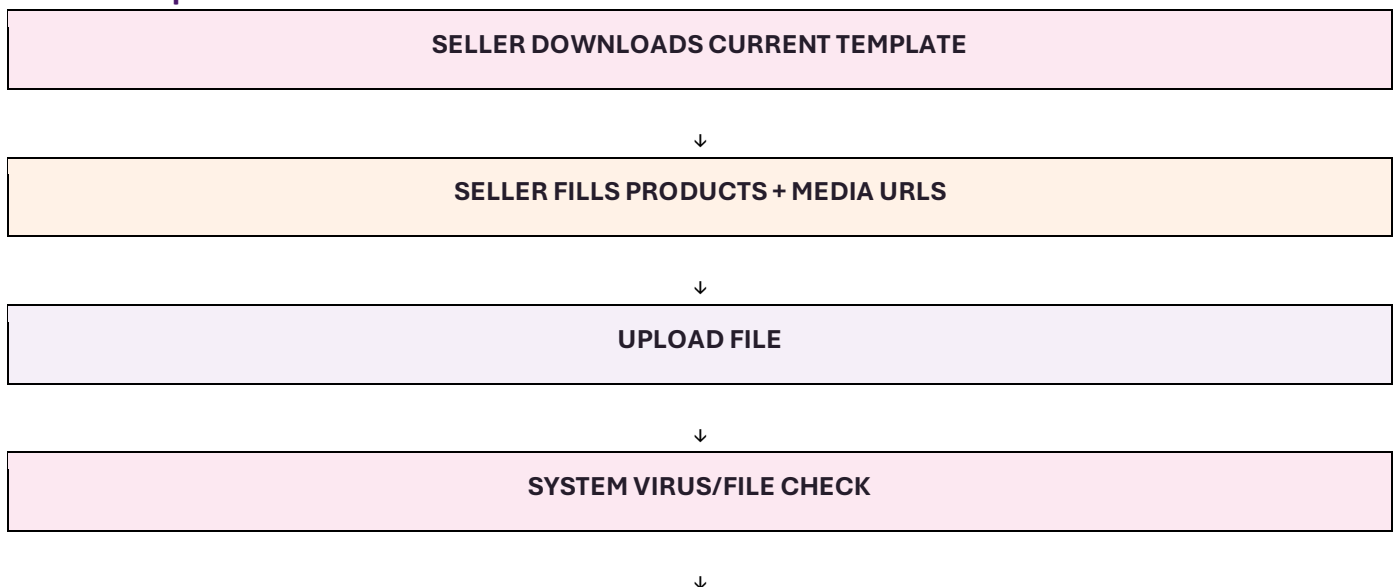
- **MUST:** Downloadable CSV template.
- **MUST:** Downloadable XLSX template.
- **MUST:** Upload completed CSV/XLSX.
- **SHOULD:** Bulk update existing products using product ID or seller SKU.
- **SHOULD:** Image import by publicly accessible image URL.
- **SHOULD:** Video URL/import field subject to media rules.
- **MAY later:** API feed / Shopify / WooCommerce connector.

8.2 Bulk Template — Minimum Columns

Column	Rule
seller_sku	Required; unique within seller account.

Column	Rule
product_title	Required.
department	Required; must map to active ISM department.
category	Required.
subcategory	Required where category requires it.
description	Required.
price	Required numeric, > 0.
sale_price	Optional; must be less than regular price when supplied.
stock_qty	Required integer for stock-controlled items.
variant_group	Optional group key for parent/variant products.
size / colour / material	As applicable to category.
weight_kg	Required for shippable products unless shipping class supplies default.
length_cm / width_cm / height_cm	Required where shipping rules demand dimensions.
handling_days	Required/defaultable.
image_1_url	Required primary image.
image_2_url ... image_n_url	Optional gallery images.
video_url	Optional media source that passes allowed-source validation.
return_eligible	Required/defaultable yes/no subject to marketplace policy.
status	draft/submitted; never allow imported rows to bypass required approval.

8.3 Bulk Import Workflow



PARSE FILE**HEADER / VERSION CHECK****ROW-BY-ROW VALIDATION****CATEGORY / ATTRIBUTE VALIDATION****DUPLICATE SKU CHECK****MEDIA URL VALIDATION****PRICE / INVENTORY / SHIPPING VALIDATION****PREVIEW: VALID / WARNING / ERROR COUNTS****SELLER CONFIRMS IMPORT****VALID ROWS IMPORTED****INVALID ROWS NOT PUBLISHED****ERROR REPORT DOWNLOAD****SELLER FIXES ONLY FAILED ROWS AND RE-UPLOADS**

8.4 Import Safety Rules

- Never partially corrupt existing data if an import fails mid-process. Use transactional/batched processing.
- Every import receives Import Batch ID, seller ID, uploader, filename, template version, start/end time and result counts.
- Import result must show: total rows, created, updated, skipped, warning, failed.
- Error report must include original row number, SKU, field, error code and human-readable message.
- Seller can cancel an import before commit; after commit, seller can view import history.
- Duplicate seller SKU must not silently create duplicate products.
- Bulk update must require explicit update mode and identify target by immutable product ID or unique seller SKU.
- Blank cell in update mode must have defined behaviour: IGNORE or CLEAR. Seller must choose; never guess.
- Admin can set maximum file size, maximum rows per import and rate limits.
- Large imports should process asynchronously with progress status and email/dashboard notification on completion.

8.5 Bulk Media Strategy

- Preferred launch approach: media URLs in CSV/XLSX so the file remains small.
- System downloads/copies allowed images into ISM-controlled media storage/CDN after validation.
- Do not hotlink seller images forever; retain a controlled copy to prevent broken listings.
- Optional future method: ZIP package containing spreadsheet + /images folder, mapped by filename.
- Media failures should fail only the affected row/asset, not the entire batch unless primary image is missing.

9. Product Media & VIDEO — CRITICAL

Video must be designed into the product model from day one. Do not bolt video on later as a free-text field.

Media Type	Launch Behaviour
Primary Image	Required; exactly one primary image.
Gallery Images	Multiple; ordered drag-and-drop.
Product Video	Seller can add video subject to limits/moderation.
Thumbnail	Generated for video or seller-selected where supported.
Alt Text	Supported for accessibility/SEO.

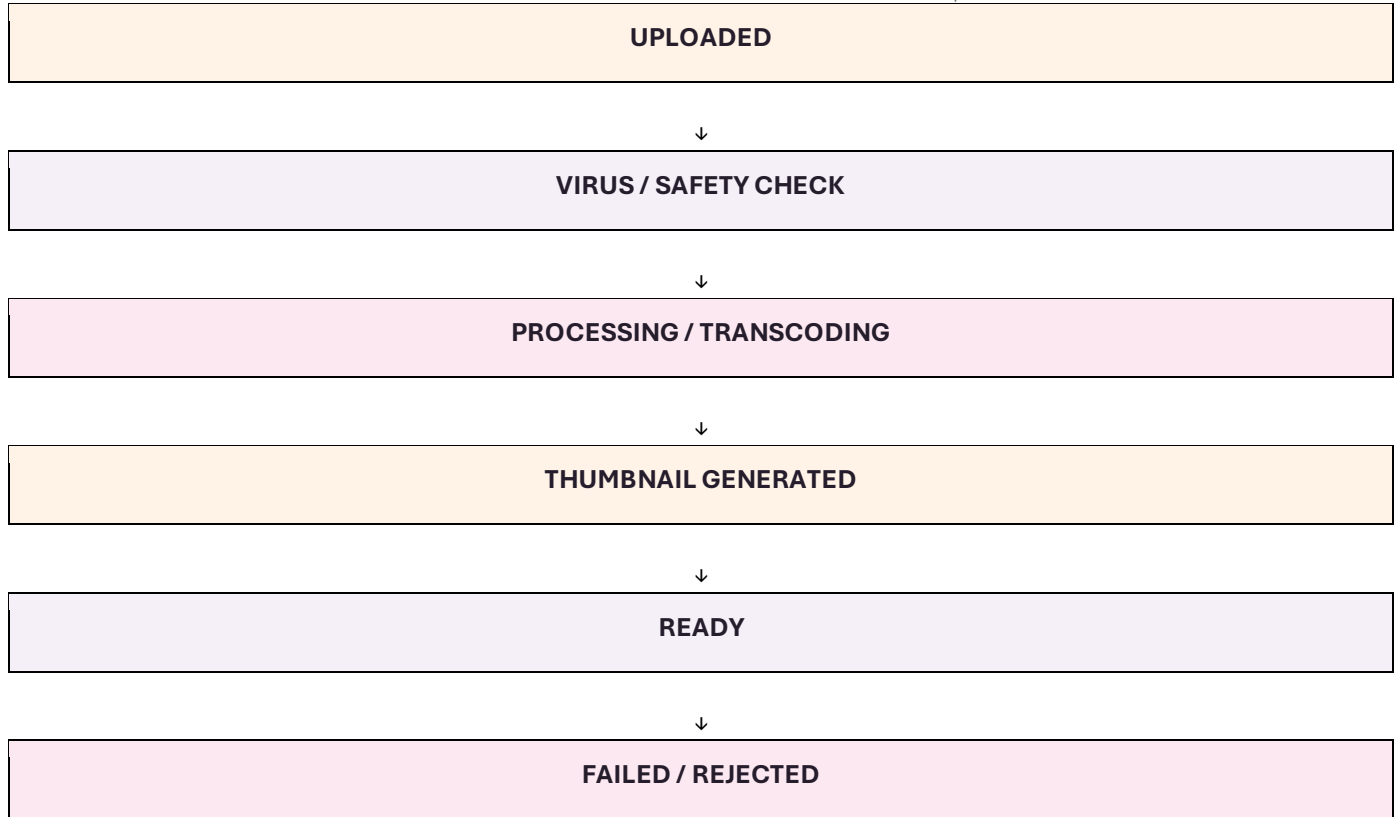
9.1 Video Input Options

- MUST support direct video upload OR a managed video-hosting integration chosen by development team.
- SHOULD support approved external video URL/embed sources only if security/privacy/performance are acceptable.
- MUST validate file type, duration, size and processing status.
- Recommended launch formats: MP4/H.264 as input; transcode/normalise via media pipeline where possible.
- Recommended product video duration: configurable, e.g. 5–60 seconds.
- Recommended seller video quantity: configurable per product, e.g. 1–3 videos.
- Video should not autoplay with sound. Muted preview may be enabled after performance testing.

9.2 Video Processing Statuses

UPLOADING





9.3 Video Failure & Moderation

- If video processing fails, product may remain Draft/Live based on admin rule, but failed video must never show broken player.
- Seller must receive clear error and be able to replace video.
- Admin can remove/reject video independently without deleting product.
- Store original upload reference and processed asset IDs; serve customer video through CDN/streaming layer.
- Video moderation status must be independent: Pending, Approved, Rejected.
- All video changes logged.

10. Variants, SKUs & Inventory

- Parent product may contain many sellable variants.
- Each sellable variant MUST support unique variant ID, seller SKU, optional barcode, price override, stock, weight/dimensions override and image mapping.
- Example: Kurta → Size S/M/L/XL × Colour Red/Blue. Each combination can carry separate stock.
- Inventory reservation must occur at checkout/payment stage to stop overselling.
- Failed/expired checkout must release reservation.
- Cancelled/refunded/returned stock behaviour must be configurable and audited.
- Low-stock alerts seller-configurable.
- Out-of-stock variant remains visible or hidden based on configuration; cannot be purchased.
- Bulk stock update must be supported separately from full catalogue upload.

11. Pricing, Tax & Promotions

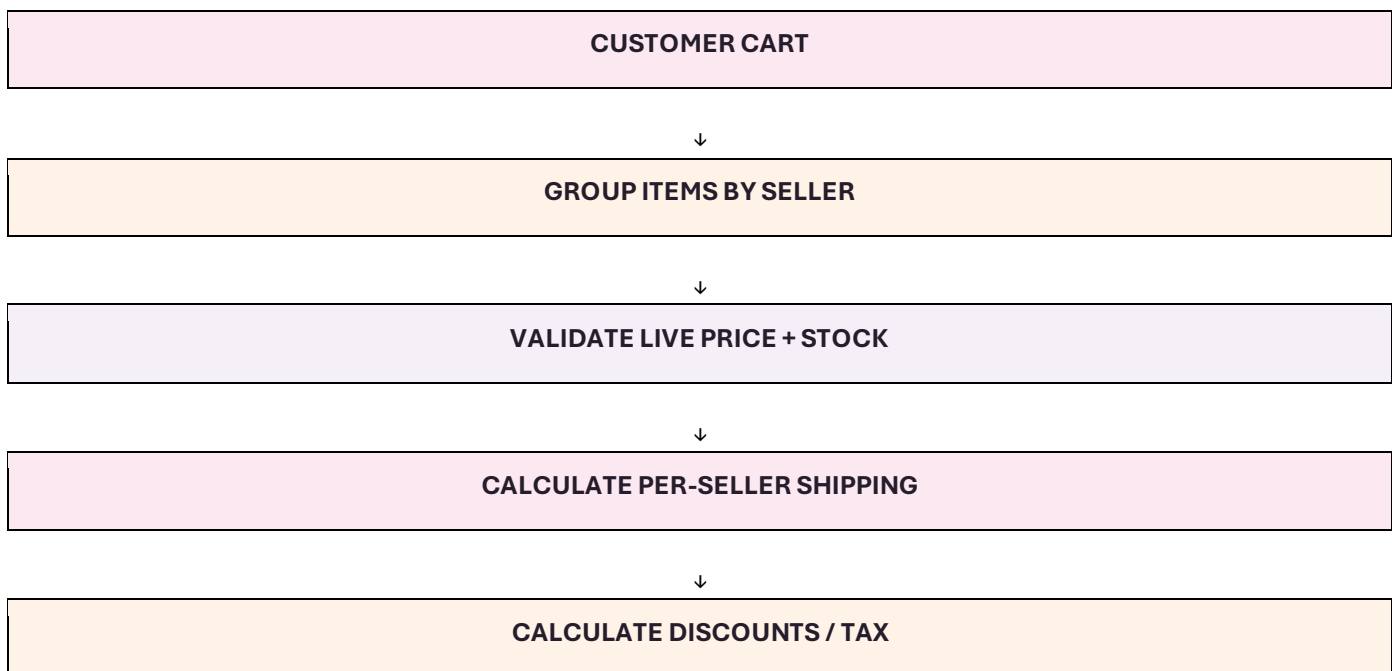
Capability	Requirement
Currency	AUD launch default.

Capability	Requirement
GST	Architecture must support GST-inclusive pricing and seller/tax reporting rules configured after accounting/legal advice.
Seller Price	Seller enters item price subject to marketplace rules.
Sale Price	Date-bound promotional price.
Marketplace Commission	Configurable globally, by seller, category or campaign if required.
Coupons	Platform coupon; seller coupon optional later.
Discount Funding	Ledger must identify whether ISM or seller funded discount.
Shipping Discount	Must identify who funds shipping promo.
Rounding	Consistent server-side monetary rounding; use decimal/money type, never floating-point for financial calculation.

12. Customer Experience

- Guest browsing and optional guest checkout.
- Account creation with email/mobile verification.
- Address book with Australian address validation where available.
- Wishlist/favourites SHOULD be included.
- Search, category browsing, filters, seller store browsing and occasion/festival discovery.
- Product page must show: title, images/video, price, variants, stock, estimated dispatch/delivery, seller, return summary and reviews.
- Customer must know when items in one cart will arrive as separate packages from separate sellers.
- Order history, tracking, returns/refunds and invoices/receipts available in My Account.

13. Cart & Multi-Seller Checkout





SHOW ONE CUSTOMER TOTAL



PAY ONCE



CREATE MASTER ORDER



CREATE SELLER SUB-ORDERS

- Cart line keeps product ID, variant ID, seller ID, quantity and price snapshot.
- Revalidate stock and price at checkout; alert customer if changed.
- Shipping can differ per seller/sub-order.
- One payment may fund multiple seller sub-orders.
- Customer-facing order number and internal seller sub-order numbers must both exist.
- Cancellation/return may occur at line/sub-order level without cancelling unrelated seller items.

14. Payments & Marketplace Ledger

Financial records must be ledger-based. Do not derive historic payouts later from editable current product prices or commission settings.

Ledger Component	Examples
Customer Charge	Items + shipping + tax - discounts
Seller Gross	Seller item amount eligible before fees/adjustments
ISM Commission	Marketplace fee
Payment Processing	If allocated to seller/ISM according to commercial model
Shipping Charge	Customer-paid shipping
Shipping Cost	Courier cost
Refund	Full/partial
Adjustment	Manual approved debit/credit
Payout	Amount sent to seller

- Every monetary entry stores currency, amount, type, order/sub-order reference, seller, timestamp and source.
- Refund and payout must be idempotent: duplicate webhook/retry cannot pay twice.
- Payment provider events must be stored and reconciled.

- Never mark order PAID solely from front-end success page; confirm server-side/provider status.

15. Order Architecture

Level	Purpose
Master Order	Customer's complete checkout.
Seller Sub-Order	One seller's portion, own fulfilment/payout.
Order Line	Specific product/variant quantity and captured price.
Shipment	One or more packages for a seller sub-order.
Return	One or more order lines returned.
Refund	Financial refund linked to return/cancellation/dispute.
Payout	Seller settlement linked to eligible ledger entries.

Required order-state model

State Group	States
Payment	PAYMENT_PENDING, PAYMENT_FAILED, PAID
Fulfilment	ORDER_CREATED, SELLER_NOTIFIED, SELLER_ACCEPTED, PREPARING, READY_TO_SHIP
Shipping	LABEL_CREATED, PICKUP_SCHEDULED, SHIPPED, IN_TRANSIT, OUT_FOR_DELIVERY, DELIVERED
Return/Refund	RETURN_REQUESTED, RETURN_APPROVED, RETURN_IN_TRANSIT, RETURN_RECEIVED, REFUND_PENDING, REFUNDED
Settlement	PAYOUT_HOLD, PAYOUT_ELIGIBLE, PAYOUT_PROCESSING, PAID_TO_SELLER
Exception	CANCELLED, DISPUTED

16. Seller Fulfilment

15. New paid seller sub-order created.
16. Seller receives dashboard + email/push/SMS as configured.
17. Seller opens order and confirms item/variant/quantity and dispatch deadline.
18. Seller accepts order. Auto-accept may be a future configurable option.
19. Stock is already reserved/deducted; seller must not modify sold quantity.
20. Seller prepares and packages item.
21. Seller confirms package weight/dimensions if actual differs and allowed.
22. Seller clicks Ready to Ship.
23. System requests shipment/label from delivery integration.
24. Seller prints/downloads label.
25. Seller chooses courier pickup or supported drop-off method.
26. Tracking events update automatically.
27. Delivery confirmation starts return and payout clocks.

Late Seller Handling

- Dashboard countdown to dispatch deadline.
- Reminder before SLA breach.
- Escalation to admin after breach.
- Customer informed if material delay.
- Repeated breaches feed seller performance score.
- Admin may cancel order if seller fails to fulfil within configured maximum period.

17. Shipping / Delivery API Architecture

Function	Required Behaviour
Rate Quote	Return available services/cost/ETA based on origin, destination, weight/dimensions.
Create Shipment	Generate courier shipment after order/ready-to-ship.
Label	Return label PDF/URL or printable asset.
Pickup	Request/track pickup where provider supports it.
Tracking	Receive polling/webhook status updates.
Cancel Shipment	Cancel unused label/shipment where supported.
Return Label	Generate return shipment where commercial rules allow.
Proof of Delivery	Store delivery timestamp and available POD metadata.

Shipping Data Hierarchy

- Origin = seller dispatch address.
- Destination = customer delivery address.
- Package = actual or catalogue package weight/dimensions.
- Service = selected courier/service.
- Cost = quoted carrier cost; customer charge may differ if ISM subsidises/free-shipping policy.
- Tracking ID and carrier shipment ID stored on shipment record.

Webhook events must map carrier-specific statuses into ISM standard statuses. Unknown carrier statuses must be stored raw and flagged, not discarded.

18. Tracking & Notifications

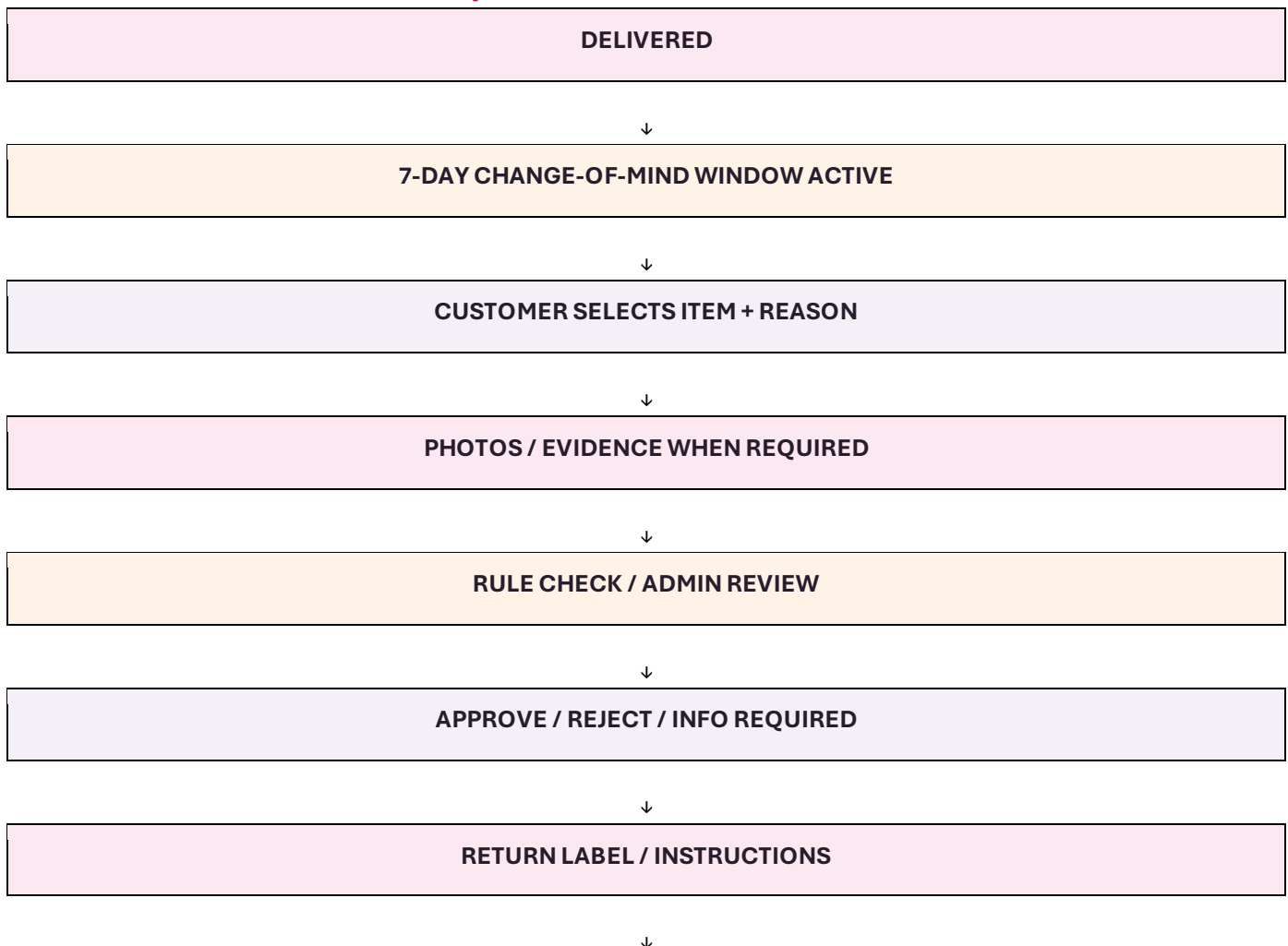
Trigger	Customer	Seller	Admin/System
Payment success	Order confirmation	New order	Create order + ledger
Seller accepted	Preparing update	Confirmation	Start SLA monitoring
Label created	Optional update	Label available	Store carrier ID
Shipped	Tracking notification	Shipment confirmed	Set SHIPPED
Out for delivery	Out-for-delivery notice	Status update	Set OUT_FOR_DELIVERY

Trigger	Customer	Seller	Admin/System
Delivered	Delivered + return/review CTA	Delivery confirmation	Start return + payout timers
Return opened	Return acknowledgment	Return request	Payout hold
Refund processed	Refund confirmation	Debit/adjustment notice	Ledger/reconciliation
Payout processed	—	Payout statement	Reconciliation

19. Cancellations

- Customer may request cancellation before shipment subject to order state and policy.
- Seller may request cancellation with reason; repeated seller cancellations affect performance.
- Admin can cancel for fraud, stock failure, prohibited product or operational reason.
- Cancellation must release inventory appropriately.
- Unused shipping label should be cancelled where supported.
- Refund must originate from the payment/ledger system, not a status toggle.
- Cancellation reason codes required for analytics.

20. Returns, Refunds & Disputes



RETURN IN TRANSIT**SELLER RECEIVES****CONDITION CONFIRMED / ESCALATED****REFUND****PAYOUT HOLD RELEASED OR ADJUSTED**

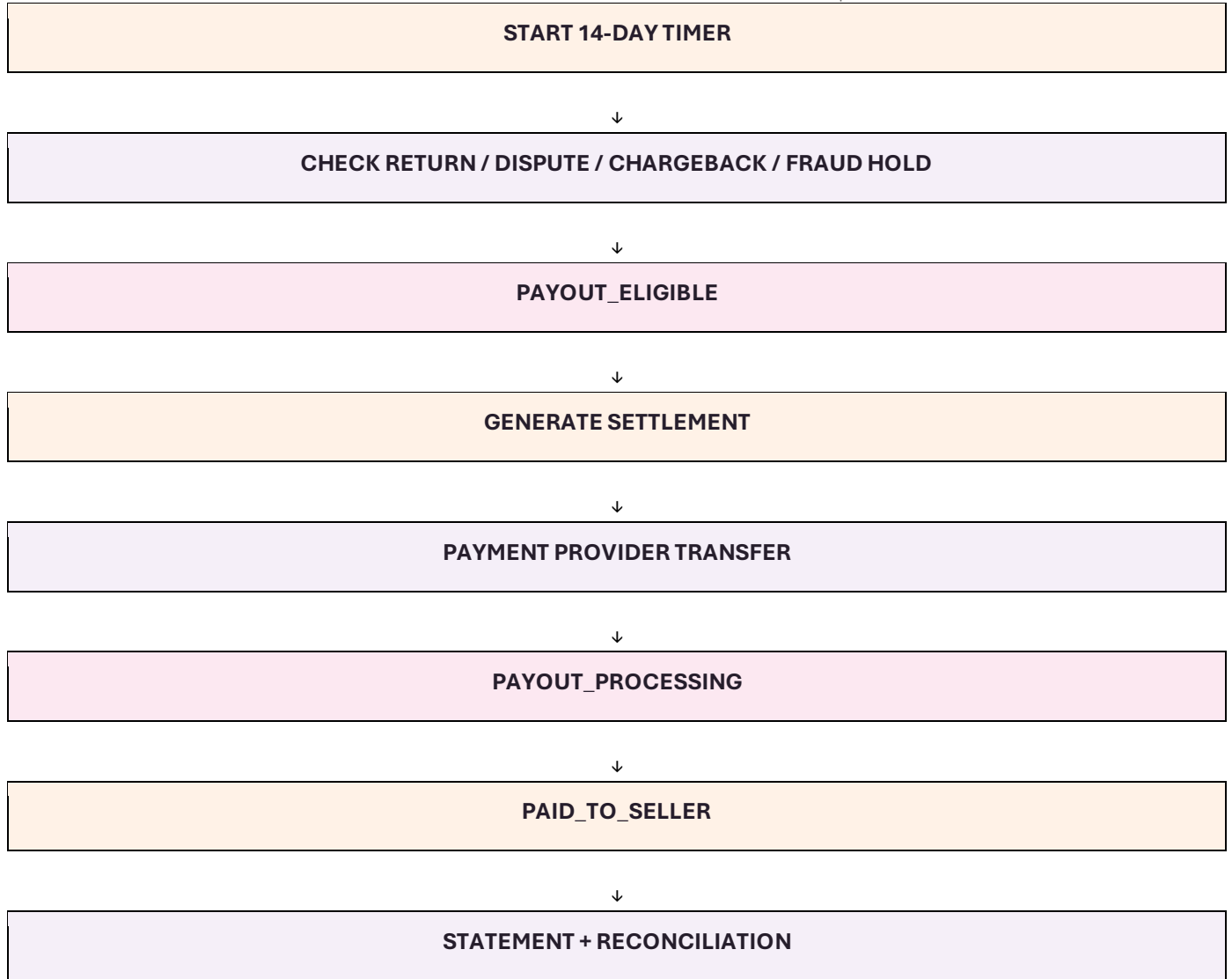
Return Reason	Typical Handling
Changed mind	Within policy window; return shipping/restocking rules configurable.
Wrong size	Policy/category dependent.
Wrong item sent	Seller fault; platform may require seller-funded return.
Damaged in transit	Evidence + courier claim path if applicable.
Defective/faulty	Consumer law workflow; must not be blocked solely by 7-day change-of-mind window.
Not as described	Evidence/review; consumer law implications.

- Return request has independent Return ID.
- Partial return of a multi-item sub-order must be supported.
- Partial refund must be supported.
- Refund may include product, shipping or both depending on reason/policy.
- Active return/dispute automatically places relevant payout amount on hold.
- Admin can add notes and evidence attachments.

21. Seller Payouts — Detailed

Payout eligibility = confirmed delivery timestamp + 14 days, provided no hold/dispute/refund blocks the amount.

DELIVERED



- Payout calculation must use captured order ledger, not current commission settings.
- Allow payout batching by seller.
- Support failed payout state and retry without duplicate payment.
- Seller dashboard shows Pending, On Hold, Eligible, Processing and Paid.
- Finance admin can place manual hold with required reason and audit log.
- Any manual financial adjustment requires reason, user and timestamp.

22. Reviews & Trust

- Only verified purchasers may review a purchased product/seller.
- Product rating and seller rating stored separately.
- Review eligibility starts after delivery.
- Admin moderation/reporting capability.
- Seller can respond to review if enabled.
- Prohibit seller self-review and duplicate order review.
- Trusted Seller badge can later use fulfilment, cancellation, return and rating thresholds.

23. Search, Filters & Discovery

- Search title, description, SKU where appropriate, category, attributes, seller and tags.

- Typo tolerance/fuzzy matching SHOULD be supported.
- Filters should be category-aware: size, colour, fabric, price, seller location, ready-to-ship, rating, occasion, festival.
- Sort: Relevance, Newest, Price Low-High, Price High-Low, Best Selling, Rating.
- Out-of-stock handling configurable.
- Search index must update when product becomes live/paused/out-of-stock/price-changed.
- Homepage collections: New Arrivals, Best Sellers, Wedding, Festival, Under \$50, Sale.

24. Admin Dashboard — Full Control

Admin Module	Must Include
Overview	GMV, orders, active sellers, products, pending approvals, returns, payout liabilities, alerts.
Sellers	Applications, verification, suspension, commission overrides, notes, performance.
Catalogue	Categories, attributes, products, moderation, bulk import monitoring.
Orders	Master/sub-orders, status, shipment, customer/seller details, timeline.
Returns	Queue, evidence, decisions, labels, refunds, disputes.
Finance	Transactions, ledger, refunds, payout holds, eligible payouts, statements, reconciliation.
Shipping	Providers, services, rate settings, shipments, tracking exceptions.
Promotions	Coupons, banners, collections, featured sellers/products.
Content	Homepage sections, static pages, policies, FAQs.
Users/Roles	Admin staff, roles, permissions, MFA requirements.
Reports	Sales, seller, product, category, returns, delivery, finance.
Audit	Searchable audit log of sensitive actions.
Settings	Commission, payout delay, return window, SLA, media limits, import limits, notification templates.

25. Seller Dashboard — Required Screens

Screen	Features
Dashboard	Sales, orders requiring action, dispatch deadlines, inventory alerts, pending payout.
Products	List/search/filter, create, edit, clone, archive, status.
Bulk Upload	Template download, new import, update import, progress, history, errors.
Inventory	Variant stock, bulk stock edit/upload, low stock.
Orders	New, preparing, ready, shipped, delivered, cancelled.

Screen	Features
Shipping	Labels, pickup/drop-off, tracking, shipment history.
Returns	Requests, evidence, action/status, return tracking.
Payments	Pending, hold, eligible, paid, statements.
Store	Logo/banner/about/policies/holiday mode.
Promotions	Seller promotions if enabled.
Reports	Sales, products, returns, fulfilment.
Team	Invite staff + permissions.
Settings	Addresses, notifications, defaults, account/security.

26. Customer Account — Required Screens

- Profile & contact verification.
- Saved addresses.
- My Orders with master order and package/sub-order view.
- Track package.
- Return/refund centre.
- Wishlist/favourites.
- Reviews.
- Notification preferences.
- Security/password/session management.

27. Notifications Engine

- Channels: in-app/dashboard, email; SMS/push optional/configurable.
- Template variables must be server-generated and escaped securely.
- Notification log stores recipient, channel, template, order/seller reference, send status and retry.
- Do not send duplicate order/payment messages on repeated provider webhooks.
- Admin can edit non-sensitive template copy without code deployment where practical.
- Critical notifications: payment success/failure, seller new order, dispatch deadline, shipped, delivered, return events, refund, payout.

28. Reporting & Analytics

Report	Minimum Metrics
Marketplace	GMV, net sales, orders, AOV, buyers, conversion where analytics supports it.
Seller	GMV, orders, acceptance, cancellations, dispatch time, returns, rating, payout.
Product	Views, sales, units, conversion, stock-outs, returns.
Category	GMV, units, sellers, products, return rate.

Report	Minimum Metrics
Shipping	Cost, average delivery time, late delivery, failed delivery.
Returns	Reason, rate, seller/category/product trends.
Finance	Customer collections, refunds, commission, shipping, liabilities, payouts.
Bulk Import	Imports, rows, error rates, common validation failures.

Exports SHOULD support CSV/XLSX for authorised users.

29. Security, Audit & Compliance

- HTTPS everywhere; secure cookies/tokens.
- Password hashing using modern standard; no plaintext passwords.
- MFA required for Super Admin and Finance Admin; recommended for sellers.
- Role-based access control enforced server-side.
- Rate limiting and bot/abuse protection on auth, checkout, uploads and APIs.
- File uploads scanned/validated; never trust file extension alone.
- Sensitive payment data handled by payment provider; minimise PCI scope.
- Do not expose full bank/payout credentials in UI/logs.
- Audit log for seller approval, product moderation, refund, payout hold/release, commission change, bank/payout change, admin role change.
- Privacy controls and data retention/deletion process must be designed for Australian requirements and legal review.
- Backups encrypted and restore-tested.

30. Performance & Reliability

- Use CDN for product images/video delivery.
- Generate responsive image sizes and WebP/AVIF where supported.
- Lazy-load below-the-fold media; video should not block page load.
- Cache category/search data appropriately without showing stale stock at checkout.
- Checkout/order/payment operations must be server-authoritative and transactional.
- Background jobs for bulk imports, media processing, emails and large exports.
- Retry external API calls with idempotency and backoff.
- Webhook handlers must be idempotent.
- Central error logging/monitoring and alerting for payment/shipping/import failures.
- Database backups and documented recovery procedure.

31. Core Data Model / Entities

Entity	Purpose / Key Fields
User	user_id, role, contact, verification, status
Seller	seller_id, owner_user_id, legal/trading/store data, status
SellerAddress	dispatch/return addresses
SellerPayoutProfile	provider account/reference, verification status

Entity	Purpose / Key Fields
SellerStaff	seller_id, user_id, permission set
Category	hierarchy, status, attributes
Attribute / Option	category-specific product data
Product	seller_id, title, category, status, content
ProductVariant	variant_id, seller SKU, options, price, stock
ProductMedia	product/variant, image/video, order, status
InventoryTransaction	variant, delta, reason, order/import reference
BulkImportBatch	seller, file, template version, counts, status
BulkImportRow	row number, SKU, validation result/errors
CustomerAddress	user/customer address
Cart / CartLine	customer/session cart data
Order	master order
SellerOrder	seller sub-order
OrderLine	product/variant snapshot
Payment	provider payment and status
LedgerEntry	financial source-of-truth entries
Shipment	seller order package/carrier/tracking
TrackingEvent	carrier raw + mapped status
Return	return request/status
ReturnLine	items/qty/reason
Refund	payment refund
Payout	seller settlement
PayoutLine	ledger entries included in payout
Review	verified product/seller review
Notification	send log
AuditLog	actor/action/entity/before-after/timestamp

32. API & Webhook Requirements

- All external integrations must have adapter/service boundaries so provider can be replaced later.
- Payment webhook: payment succeeded/failed/refunded/dispute/chargeback where provider supports.
- Shipping webhook: shipment created, picked up, in transit, out for delivery, delivered, exception, return.

- Media callback: processing ready/failed if external video service used.
- Webhook signatures must be verified.
- Webhook raw payload, provider event ID, received timestamp and processing result should be logged.
- Provider event IDs/idempotency keys must prevent duplicate business actions.
- Admin should be able to retry failed integration jobs safely.

33. Error Handling & Edge Cases

Scenario	Required System Behaviour
Customer pays but browser closes	Server/provider webhook still creates/updates paid order. Customer can retrieve from account/email.
Payment succeeds, order creation initially fails	Retry using idempotent payment reference; never charge twice.
Two buyers purchase final unit	Atomic inventory reservation; one succeeds, other receives out-of-stock response/refund protection.
Seller does not accept order	Reminder/escalation then admin cancellation per SLA.
Shipping API unavailable	Queue/retry; seller/admin sees integration exception. Do not lose order.
Label created twice	Idempotency prevents duplicate billable shipment.
Courier says delivered but customer disputes	Create delivery dispute; payout hold can be applied.
Partial delivery	Shipment-level states; seller sub-order may contain multiple shipments.
Return opened on day 7	Use delivery timestamp and configured timezone/server rule consistently.
Return still active on day 14	Payout held for affected amount.
Bulk file contains 1,000 rows and 25 invalid	975 valid rows can import only after seller preview/confirm; 25 remain failed with report.
Bulk update has blank price	Follow explicit import mode; never infer clear vs ignore.
Primary image URL broken	Row fails validation; product must not publish without primary image.
Video processing fails	Product remains usable according to rule; failed video hidden and seller notified.
Seller changes bank account before payout	Verification/security workflow + audit; pending payout rule defined.
Refund after seller already paid	Create seller negative balance/recovery adjustment according to agreed commercial/legal process.

34. Testing & Acceptance Criteria

The developer must provide evidence for these test groups before launch.

Test Group	Minimum Acceptance
Seller Onboarding	New seller can register, verify, submit, be approved, access store and dashboard.
Single Listing	Product with variants/media can be drafted, validated, approved and purchased.

Test Group	Minimum Acceptance
Bulk Upload	At least 1,000-row test file; valid import, invalid-row report, re-upload and update mode all proven.
Video	Upload/process/playback/failure/replacement/moderation tested on mobile and desktop.
Inventory	Concurrent last-unit purchase does not oversell.
Multi-Seller Cart	One checkout creates correct master + seller sub-orders and financial splits.
Payment	Success, failure, retry and duplicate webhook are idempotent.
Shipping	Rate, shipment, label, tracking and delivery webhook tested in sandbox/provider test.
Returns	Partial and full return; payout hold; refund.
Payout	14-day eligibility calculation, holds, failed payout retry, statement.
Permissions	Seller cannot access another seller's data; admin roles restricted.
Security	Auth/rate/upload validation and secrets handling reviewed.
Performance	Large category/product media pages and bulk jobs do not block checkout.

Mandatory UAT Scenarios

28. Seller imports 500 products with images and a subset of videos.
29. Customer purchases items from three sellers in one checkout.
30. One seller ships on time; one late; one cancels.
31. One order is returned within 7 days.
32. One product has a statutory fault complaint after the ordinary change-of-mind window.
33. One payout becomes eligible exactly 14 days after confirmed delivery.
34. One return opened before payout prevents that affected amount from being released.
35. Duplicate payment and shipping webhooks do not create duplicate orders, labels, refunds or payouts.

35. Launch Readiness Checklist

- ☐ Seller onboarding approved end-to-end
- ☐ Admin seller approval/rejection/info-required
- ☐ Category tree + attributes configured
- ☐ Single product listing
- ☐ Bulk CSV upload
- ☐ Bulk XLSX upload
- ☐ Bulk update
- ☐ Bulk error report and import history
- ☐ Images stored/served through controlled media pipeline
- ☐ Product video upload/processing/playback
- ☐ Variant/SKU stock management
- ☐ Low stock / out-of-stock

- ☐ Customer search/category/filter
- ☐ Cart + one checkout
- ☐ Multi-seller order split
- ☐ Payment provider live configuration
- ☐ Order/ledger reconciliation
- ☐ Seller fulfilment screens
- ☐ Shipping provider live configuration
- ☐ Label creation + tracking
- ☐ Delivery event
- ☐ 7-day return window logic
- ☐ Return/refund workflow
- ☐ 14-day payout eligibility
- ☐ Payout holds
- ☐ Seller payout statements
- ☐ Admin finance tools
- ☐ Email/SMS templates as applicable
- ☐ Audit logs
- ☐ Backup/restore
- ☐ Error monitoring
- ☐ Privacy/terms/returns/seller policies approved
- ☐ Mobile responsive QA

36. Phase 2 / Future-Ready Architecture

- Seller API and inventory feeds.
- Shopify/WooCommerce seller import connectors.
- Advanced promoted listings / sponsored products.
- Seller subscription tiers.
- Marketplace wallet/store credit.
- Gift cards.
- Bundle products and gift hampers.
- Click & collect / local pickup.
- Same-day/local delivery integrations.
- Advanced fraud/risk scoring.
- AI-assisted product description/category mapping, but never bypass required seller/admin validation.
- Mobile app using the same core APIs.
- Loyalty/rewards.
- Customer chat with privacy controls.
- Cross-border selling only if a separate compliance/logistics design is approved.

FINAL DEVELOPER RULE: When a requirement affects money, inventory, order ownership, shipping, returns, seller payouts or customer rights, it must be represented in the backend data model and state logic — not only as front-end text.

Developer Sign-off Questions

#	Developer must answer before build sign-off
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#	Developer must answer before build sign-off
1	Confirm architecture supports master order + seller sub-orders.
2	Confirm bulk CSV/XLSX create AND update flows, error report, import history and 1,000+ row processing.
3	Confirm product video architecture, storage/streaming, processing, moderation and failure states.
4	Confirm seller-level/variant-level inventory reservation prevents overselling.
5	Confirm payment and seller payout are ledger-based and idempotent.
6	Confirm shipping provider is replaceable through an adapter/integration layer.
7	Confirm delivery timestamp is authoritative for return/payout clocks.
8	Confirm return/dispute holds prevent affected seller payout.
9	Confirm role/permission isolation between sellers.
10	Confirm audit trail for sensitive financial/admin changes.
11	Confirm all external webhooks are signature-verified and idempotent.
12	Confirm launch acceptance tests will be demonstrated before production go-live.